HSC EXAMINATION 2025 - MODEL TEST PAPER

BIOLOGY FIRST PAPER (BOTANY) - CREATIVE & MCQ

1. Sample Creative Question 1 (Cell Structure)

Stimulus: A double membrane-bound cell organelle contains flattened thylakoid discs stacked into grana and stroma fluid with circular DNA.

Questions:

a) What is a ribosome? (1 Mark)

b) Explain why mitochondria are considered semi-autonomous organelles. (2 Marks)

c) Explain the ultrastructure of the organelle mentioned in the stimulus with a diagram. (3 Marks)

d) Evaluate the physiological importance of this organelle in sustaining the biosphere. (4 Marks)

2. Sample Creative Question 2 (Biotechnology)

Stimulus: Scientists transferred the cry1Ac endotoxin gene from Bacillus thuringiensis into Gossypium hirsutum to resist bollworm infestation.

Questions:

a) What is a plasmid? (1 Mark)

b) Distinguish between cohesive and blunt ends generated by restriction endonucleases. (2 Marks)

c) Describe the operational steps required to produce the transgenic organism in the stimulus. (3 Marks)

d) Analyze the ecological benefits and biosafety concerns associated with this technology. (4 Marks)

3. Selected High-Probability MCQs (2025 Series)

1. Which organelle is called the 'suicide bag' of the cell?
   (A) Peroxisome (B) Lysosome (C) Centrosome (D) Ribosome
   Answer: (B) Lysosome contains hydrolytic acid hydrolases.

2. In which sub-stage of Prophase I does genetic crossing over take place?
   (A) Leptotene (B) Zygotene (C) Pachytene (D) Diplotene
   Answer: (C) Pachytene stage forms chiasmata.

3. What is the primary carbon dioxide acceptor in C4 plants?
   (A) RuBP (B) PEP (C) Oxaloacetate (D) 3-PGA
   Answer: (B) Phosphoenolpyruvate (PEP).

4. Which bacterium is widely employed as a natural genetic engineer in plant transgenesis?
   (A) Escherichia coli (B) Agrobacterium tumefaciens (C) Bacillus thuringiensis (D) Rhizobium leguminosarum
   Answer: (B) Agrobacterium tumefaciens via Ti plasmid.